

Table 10.2 Calculation of measurement uncertainty for different batches.

Sample	Average $\bar{Z}_1^*$	Average $\bar{Z}_2^*$	s_r^2 $\times 10^{-5}$	s_B^2 $\times 10^{-5}$	$u^2(\bar{Z})$ $\times 10^{-5}$	$U(\bar{Z})$ $\times 10^{-3}$	UR % (%)
FTC-A	0.083	0.084	1.529	0.055	1.047	6.5	7.7
FTC-B	0.082	0.083	0.639	0.044	0.448	4.2	5.1
FTC-C	0.080	0.078	4.643	0.140	3.165	11.3	14.4
TDF-D	0.098	0.098	2.329	0.000	1.552	7.9	8.0
TDF-E	0.094	0.100	2.294	1.540	2.299	9.6	9.6
TDF-F	0.097	0.100	4.522	0.392	3.210	11.3	11.3

Sample	Predicted concentration T (mg/tablet)	$U(T)$	Coverage interval of T	
FTC-A	209.6	16.1	194	226
FTC-B	205.5	10.5	195	216
FTC-C	197.8	28.4	169	226
TDF-D	245.6	19.7	226	265
TDF-E	242.6	23.3	219	266
TDF-F	246.5	27.9	219	274

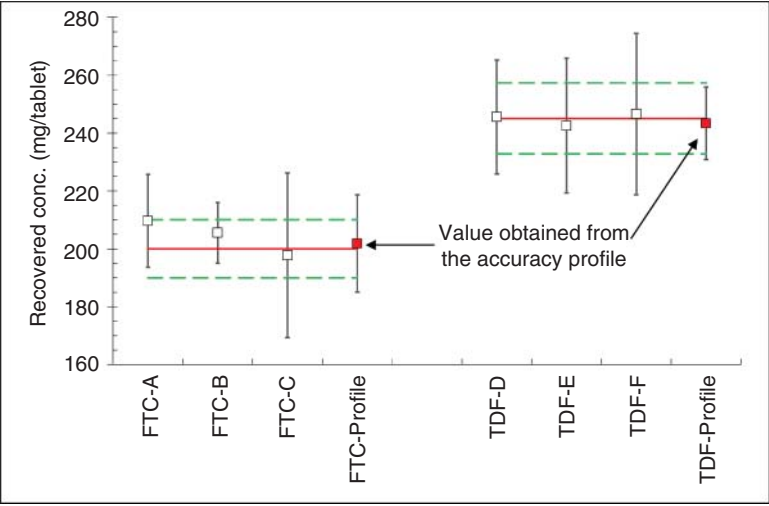


Figure 10.4 MU estimates of six drug lots for two nominal strengths of 200 and 245 mg/tablet. The error bars represent the coverage intervals. On the same graph, the accuracy profile results are plotted as a red solid squares. Same legend as Figure 10.3.

The control of TDF (tenofovir disoproxil) thus appears to be somewhat more effective than that of FTC (emtricitabine). Considering the measurements as two series and taking into account the between-series variance appears to significantly reduce the MU on the average predicted content. The new values are obtained by combining 12 absorbance measurements (instead of 9), which is relatively small. In Figure 10.4 the results obtained by applying the procedure derived from the accuracy profile are also reported as a black square, labeled FTC-Profile and TDF-Profile. The following chapter presents other possible data processings.

10.1.3 Estimation from Method Accuracy Profile

To confirm the usefulness of MU estimates obtained with HPSAM, it was decided to implement an experimental design corresponding to that recommended for obtaining a MAP using synthetic samples of known concentration. But such validation materials with known assigned concentrations were not available; therefore, a validation material was prepared by crushing and mixing 20 tablets of the drug, said to contain a mixture of 245 mg TDF and 200 mg FTC per tablet.

From the homogenized batch, independent determinations of the two APIs were performed with the classical SAM procedure, i.e. by making two additions of the same stock solution, as described above. In this case, the measurements were performed at a single wavelength of 200 nm. The experimental design itself consisted of three series of six replicates per series. Table 10.3 summarizes this dataset called ARV for antiretroviral drugs, while Table 10.4 indicates the inverse-predicted concentrations of TDF and FTC.

Table 10.3 ARV – description of the ARV dataset.

Title	ARV (antiretroviral drugs)
Reference article	[4]
Measurand	Percentage recovery of active ingredient, expressed as a function of the expected nominal content
Method	Zone capillary electrophoresis with UV detection at multiple wavelengths.
Acceptance interval	$\pm 5\%$. Value classically used for fraudulent drug analysis
Validation materials	Mixture of 20 crushed tablets
Validation plan	Number of series ($I = 3$); number of replicates ($J = 6/\text{series}$); number of levels or products ($K = 2$)
Calibration plan	See the reference article.
Total number of measures	18 measurements on each of the API present, tenofovir (TDF) and emtricitabine (FTC) in a surrogate sample. Each of these measurements is made at a single wavelength with 2 standard additions and a zero addition, for a total of 108 measurements.
Inverse-predicted concentrations	Table 10.4

Table 10.4 ARV – experimental design and levels found for the two molecules.

Replicates	Tenofovir disoproxil (TDF) 245 mg/tablet			Emtricitabine (FTC) 200 mg/tablet		
	Series 1	Series 2	Series 3	Series 1	Series 2	Series 3
1	239.1	251.9	245.7	214.9	205.7	194.9
2	243.5	244.9	233.7	191.2	201.6	209.3
3	240.5	245.8	234.5	202.4	203.9	199.5
4	250.4	244.7	239.3	215.7	192.0	213.5
5	244.5	239.6	242.8	192.3	194.8	200.0
6	250.4	251.6	237.3	205.7	190.5	205.2

Table 10.5 ARV – estimated MU and coverage interval from MAP.

Parameter	Symbol	TDF	FTC
Average predicted content	T	243.3	201.8
Repeatability standard deviation of	s_r	4.7262	8.2027
Between-series standard deviation	s_B	3.4513	0.0
Intermediate precision std. dev.	s_{FI}	5.8522	8.2027
Effective measures	N_E	8.07	16.62
Coverage probability	β	95%	95%
Coverage factor	k_{IT}	2.30	2.11
Coverage interval		228.9	184.0
		257.8	219.7
Expanded uncertainty	$U(T)$	12.6	16.9
Relative uncertainty	$UR\%(T)$	5.16%	8.35%

From these data, various validation parameters are calculated using the Resource Q worksheet. The main results are reported in Table 10.5. Unlike other accuracy profiles in this book, results are not expressed as recovery yields referring to the assigned nominal content. This mode of expression cannot be used because the tablet's content is unknown; it is only assumed.

The proportion β applied for these calculations is 95% since the goal is to evaluate the 95% expanded uncertainty. This means that 95% of the probable values of the inverse-predicted concentrations lie in the interval [229; 258] mg/tablet for an assumed level of 245 mg of TDF and between [184; 220] at 200 mg/tablet of FTC. Because the content announced by the manufacturer of the imported drug is included within the estimated coverage interval, the product can be considered conforming. The coverage intervals depend on the relative uncertainty, which is about 5% for TDF and 8% for FTC at this concentration. The two values make it impossible to reach an acceptance interval of $\pm 5\%$.

If the products are considered conform, the recovery yields apply to the nominal content of the drug products. Thus, the coverage intervals of the recovery yields are [93%; 105%] for TDF and [92%; 110%] for FTC, respectively. They can be considered satisfactory since the value of 100% is included in the coverage intervals.

This comment highlights that a decision based on a coverage interval is not exactly a statistical null hypothesis test, although the conclusion is comparable. It should be remembered that for the buyers' club, it is impossible to have validation materials with exactly known content. The value claimed by the manufacturer, is considered here as a true value. However the potential bias of the method should be corrected using the standard addition method. This does not pose any problem for estimating the MU; on the contrary, since the GUM recommends updating all significant sources of bias before estimating the MU.

For FTC, the main source of uncertainty is entirely due to repeatability with a value of 4.1%; the between-series effect (in this case, inter-day) is zero. In the case of TDF, while the MU is smaller, the repeatability and the between-series effect are approximately equivalent.

When graphically comparing the various approaches (Figures 10.3 and 10.4), the HPSAM, repeated in two series, is adequate if it is required that the acceptance interval must be within $\pm 10\%$. This can easily be explained by a more significant number of degrees of freedom than when only one measurement is used. In this case, the risk of a false decision in releasing a batch of imported drugs is not negligible in the context of the buyers' clubs.

10.2 Method Comparison Using Uncertainty

10.2.1 Analyte Defined by the Operating Procedure

As previously discussed, some analytes are defined *per se*, i.e. by the analytical operating procedure. This usually means that their exact chemical formula, in terms of clearly identified molecular composition, is not fully established. There are many examples, such as blood cholesterol, moisture in various forms and matrices, dietary fiber, protein, total organic carbon, and oil acidity. It could be considered that the standard operating procedure defines the analyte.

These analytes were often introduced a long time ago with old methods and, over time, have become references for medical diagnosis, technical control, or commercial exchange. For example in Europe, the total fat content of raw milk is used for payment for quality. The AOAC and FDA have taken to referring to these as *gold standard* methods. It seems preferable to refer to them as official or reference methods, as gold has no longer been a monetary standard since 1971 under the presidency of Richard Nixon.

The classic disadvantages of traditional methods are that they are time-consuming, labor-intensive, and often require hazardous reagents; they are therefore expensive, and many publications suggest replacing them with alternative methods that are faster and cheaper.

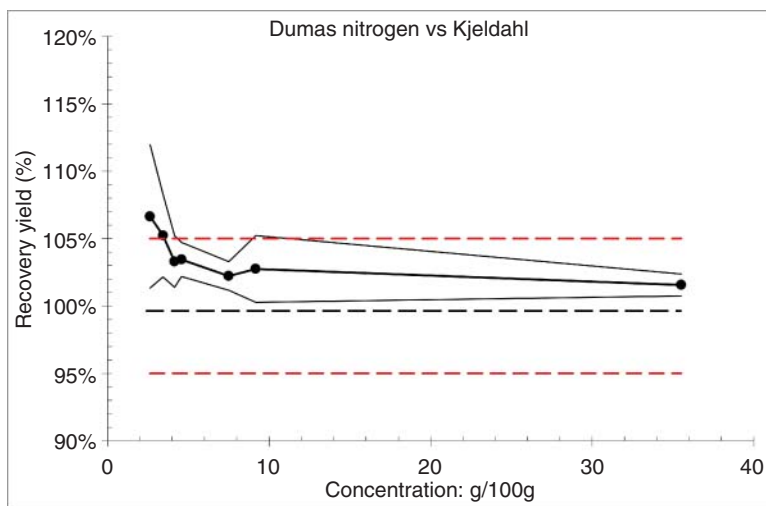


Figure 10.5 Accuracy profile of Dumas's method applied to dairy products using Kjeldahl method as reference. β -ETI with $\beta = 80\%$. Acceptability $\pm 5\%$.

The so-called Kjeldahl nitrogen is one of the official methods for measuring total proteins in foods and feedstuffs. The original procedure was developed at the end of the nineteenth century by a Danish chemist. It consists of an acid attack of the test portion by concentrated sulfuric acid, boiling at reflux for hours. This action is supposed to destroy the molecules containing organic nitrogen (proteins and nucleic acids) and mineralize them into ammonium sulfate.

Since its invention, even if there are many variants, the principle has remained the same, and the damage due to acid-generated vapors is devastating for laboratories. Finally, the conversion of Kjeldahl nitrogen into total proteins is done by multiplying the result by a consensus factor that depends on the matrix type.

The Dumas method, on the other hand, is not much newer, but it has at least two advantages: speed and the absence of strong acid. It consists of total oxidation of nitrogen by combustion in the presence of excess oxygen. For a long time, it was difficult to apply, but today it is combined with an efficient catharometric determination of nitrogen in gaseous form after the elimination of sulfur and carbon oxides and a reduction of nitrogen oxides to gaseous nitrogen.

A few years ago, these two methods were automated by various manufacturers. For various foodstuffs, such as dairy and cereal products, the question arose whether the official Kjeldahl method could be replaced by the faster and less expensive Dumas' method. Several validation studies were carried out. Figure 10.5 presents one of these studies for dairy products using the MAP¹.

¹ Unpublished personal data.